NOTATION · CHEAT SHEET

Integral Notation at a Glance

The marks from the notation cards across the integral pages: how each is read and what it means. Every group links to the full card on its page.

INTEGRALS

$\int$

THE INTEGRAL OF
A stretched S, for sum.

$f(x)$

THE INTEGRAND
The function being integrated.

dx

D X
Names the variable of integration; closes the expression.

DEFINITE INTEGRALS

$\int_a^b f(x)\,dx$

THE INTEGRAL FROM A TO B
A number: the signed area from a to b.

$\int_0^1 t^2\,dt$

A DUMMY VARIABLE
Equals $\int_0^1 x^2\,dx$: the letter does not matter.

$\Big[F(x)\Big]_a^b$

F OF X, EVALUATED FROM A TO B
$F(b) - F(a)$; also written $F(x)\,\big|_a^b$.

$\sum f(x_i^*)\,\Delta x$

THE RIEMANN SUM
Its limit is the definite integral.

INDEFINITE INTEGRALS

$\int f(x)\,dx$

THE INDEFINITE INTEGRAL OF F
A family of functions, not a number.

$F(x) + C$

F OF X PLUS C
Every antiderivative; C is any constant.

$F'(x) = f(x)$

F PRIME EQUALS F
What makes F an antiderivative of f.

IMPROPER INTEGRALS

$\int_a^\infty f(x)\,dx$

THE INTEGRAL FROM A TO INFINITY
Shorthand for a limit of integrals.

$\lim_{t\to b^-} \int_a^t$

LIMIT AS T APPROACHES B FROM BELOW
How a bad endpoint is handled.

$\int < \infty$

THE INTEGRAL CONVERGES
The limit exists and is finite; otherwise it diverges.

INTEGRATION TECHNIQUES

$u = g(x)$

LET U EQUAL G OF X
Substitution: with $du = g'(x)\,dx$, rewrite the whole integral in u.

$\int u\,dv = uv - \int v\,du$

INTEGRATION BY PARTS
Trade one integral for another.

$x = a\sin\theta$

LET X EQUAL A SINE THETA
Trigonometric substitution for $\sqrt{a^2 - x^2}$.

SAME MARKS, DIFFERENT JOB

$\int$ with vs without limits	With limits, a number; without, a family of functions plus C.
dx	Part of the integral sign here; a differential on the derivative pages.
Δx vs dx	A finite width in a Riemann sum; in the integral, the width has gone to zero.